

GENERAL SPECIFICATION G6.17

PLC Hardware (for other than C5A)

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G6.17 PLC HARDWARE (FOR OTHER THAN C5A)

G6.17.1 General

- (a) Programmable logic controllers (PLCs) shall control the plant via discrete (4-20mA HART and 24 Vdc) input and output modules and via Ethernet interface to motor control centres (MCC's) and variable speed drives (VSD's).
- (b) PLCs shall be classified as either "main plant PLCs" or "machine PLCs".
- (c) Main plant PLCs shall be those shown in the Drawings and any others required to meet the Specification. Machine PLC's shall only be permitted for complex machines, in all other instances process equipment shall be controlled by main plant PLC's. Complex machines shall be as defined in Particular Specifications P6, or where not defined in P6 shall be:
 - (i) Centrifuges;
 - (ii) Bio gas flares and boilers;
 - (iii) Air compressors;
 - (iv) Aeration Air blowers;
 - (v) Hydraulic Power Units; and
 - (vi) UV disinfection units.
- (d) Machine PLC's shall only be used to control the machines themselves and their essential ancillaries, but not associated process equipment. Under exceptional circumstances, auxiliary instrumentation related to the machine process could be connected to the machine PLC upon S.O. approval.
- (e) Machine PLC's shall use 24VDC control voltage, 4-20mA.
- (f) Vendor packages and design & build facilities shall be controlled by hot-standby PLC's which shall comply with the requirements for main plant PLC's. Refer to Particular Specification P6 ICA for these requirements.
- (g) Vendor packages and Design & Build Facilities shall be defined as:
 - (i) Thermal Hydrolysis Process;
 - (ii) Wet Weather Facility; and
 - (iii) Sidestream N removal facility.
- (h) Apart from for complex machines, all instrumentation, valves, motor starters and drives shall be connected directly to the appropriate main plant PLCs as shown in the Drawings and not via any other PLC.
- (i) All PLC hardware shall have conformally coated circuit boards. The conformal coatings shall be applied within the PLC hardware manufacturer's supply chain and shall be approximately 25-50 um thick polymeric films which cover and encapsulate the printed circuit assemblies.
- (j) All PLC hardware shall be capable of operating in a harsh and corrosive industrial environment without requirement for air-conditioning or power supply filtering.
- (k) Main plant PLC hardware shall be protected from voltage surges as specified in General Specification G6.2 Installation Requirements.

G6.17.2 Machine PLCs

- (a) Machine PLCs shall be those controllers supplied as standard original equipment by manufacturers of complex machines. Machine PLCs must be specifically listed in the Technical Schedules. In all other instances equipment shall be controlled by main plant PLC's.
- (b) Machines may incorporate PLCs of a different brand or model-family to the main plant PLCs.
- (c) All machine PLCs shall be locally supported directly by the manufacturers' Singapore subsidiary branch office, not via a local agent or distributor. The models offered shall have a wide local installation base consisting of at least 100 installations in Singapore across any industries.
- (d) Machine PLCs shall not form a single point of failure in terms of operation of the treatment plant. The failure of any one machine PLC or component thereof shall not prevent the treatment plant from continuing to operate under automatic control, including operating at maximum capacity and meeting all performance parameters.
- (e) Where machine systems consist of a number of individual machine PLC's and one overall master PLC for coordinating the individual machines, then the master machine PLC shall have hot-standby CPU's, dual power supply units, dual power supply feeds from UPS distribution board, multiple communication links to the plant automation network as shown on the Drawings.
- (f) Machine PLCs for individual machine units shall preferably be interfaced the dual ring PAN using a data communications link, but interfaces may use discrete i/o signals with the S.O.'s approval. Data communications links shall be Ethernet-based using a widely-supported open standard data communication protocol, achieving direct peer-to-peer communication without the use of intermediate protocol converters. Each individual machine PLC interfaced by data communications link shall have a link to each of the dual rings forming the plant automation network (PAN).
- (g) Machine PLC's controlling multiple machine units shall be interfaced with main plant PLC's using a data communications link. Data communications links shall be Ethernet-based using a widely-supported open standard data communication protocol, achieving direct peer-to-peer communication with the plant PLC without the use of intermediate protocol converters. Each of the master machine PLC duty and standby processors, shall have two communication links to the dual ring plant automation network (PAN). Thus there will be four network connections between a hot-standby master machine PLC and the PAN.
- (h) Where machine PLCs are connected by data communications, the machine PLC shall provide full access to the SCADA system for display of the status of components within the machine and to provide alarming and trending of machine internal parameters. Facilities for remote control (as opposed to display of status) will depend on the General Specifications G4 and may be more limited than for plant that is connected directly to main plant PLCs, however remote-control facilities shall include starting, stopping, duty selection and adjustment of key setpoints.
- (i) Program and data memory shall be provided with a minimum of six months integral battery or capacitor back-up. It shall be possible to replace any backup battery without memory loss.

- (j) Machine PLCs shall contain sufficient communication ports such that a portable programming device can be connected and operated locally without disconnecting the PLC from any permanent network.
- (k) Communication failure watchdogs shall be configured.
- (l) All Machine PLC hardware shall have conformally coated circuit boards. The conformal coatings shall be applied within the PLC hardware manufacturer's supply chain and shall be approximately 25-50 um thick polymeric films which cover and encapsulate the printed circuit assemblies.

G6.17.3 Main Plant PLC Input / Output

- (a) Conventional I/O with HART shall be used:
 - (i) PLCs shall be fitted with input and output circuits to interface with field devices for data acquisition and control. Input and output circuits shall be arranged on plug-in modules to facilitate replacement. Plug-in input and output modules shall be fitted with connectors which allow replacement of the module without disconnection of field interface wiring. All field inputs, both analogue and digital, shall be isolated and shall not be affected by differing common mode potentials.
 - (ii) Inputs and outputs for any set of duty/standby equipment shall be arranged over multiple I/O cards such that even after the failure of a single I/O card, operation and monitoring by the PLC of at least 50% of the units shall still be possible.
 - (iii) When commissioning is complete, PLCs shall have a minimum, at each i/o panel, of 20% installed spare digital inputs and outputs, and 30% spare analogue inputs and outputs, all wired down to terminals. Input and output capacity shall be further expandable in that there shall be sufficient spare slots to allow the installation of an additional 30% of I/O cards. The spare slot requirement shall be subject to a minimum of 4 spare slots per panel.
 - (iv) The Contractor shall be responsible for considering each output load in detail to ensure that the current capabilities of the PLC output channel are adequate.
 - (v) Analogue input modules shall be 4-20 mA HART modules using differential mode inputs. HART data shall be passed through the PLC to SCADA.
- (b) Connections to Field Devices
 - (i) Instruments and valve actuators shall be connected to the PLC using conventional 4-20mA HART and 24 Vdc discrete signals.

- (c) I/O Allocation
 - (i) I/O shall be allocated in a manner such that the failure of any single I/O module cannot disable more than one unit of plant. For example, such a failure shall not disable more than either:
 - 1) One process train or;
 - 2) One blower or;
 - 3) One pump (per pump station) or;
 - 4) One VSD or;
 - 5) One chemical dosing unit, etc.
 - (ii) No single failure shall prevent the automatic operation of both the duty and standby unit for any equipment.
- (d) P&ID Drawings for local control station types shall define required interface signals for valve actuators.

G6.17.4 Interface with Switchboards and Motor Control Centres

- (a) Interface for Motor Control
 - (i) P&ID Drawings for local control station types define the required interface signals for motors.
 - (ii) Interface between Main Plant PLCs and motor starters, including variable speed drives, shall be by Ethernet based field network as shown in the Drawings.
 - (iii) MCC's shall have data concentrators provided by the MCC manufacturer, and thus the interface of motors to the field network shall be via the Data Concentrators.
- (b) Interface for Electrical Switchgear
 - (i) Any digital input and output signals between PLCs and switchboards (other than for motor starters) shall be 24 Vdc signals. Each digital output from a PLC shall provide 24 Vdc to energise the coil of an interposing relay located in the respective switchboard cubicle. Digital inputs to PLCs shall be powered from the PLC end and switched by a potential free contact in the switchboard cubicle. Analogue signal loops shall be arranged to suit the relevant equipment and as such may be powered from either end.
 - (ii) Refer to Particular Specifications P6 for the required interface signals.
 - (iii) The Contractor shall provide any other interface signals that are required to comply with other sections of the Specifications.

G6.17.5 Main Plant PLC Programming

- (a) Main plant PLC programming is by others except where stated otherwise in Particular Specifications P6 ICA.
- (b) PLC Programming for Vendor Package Plants and for Design & Build Plants shall be by the Manufacturer or Contractor. Refer to Particular Specifications.

G6.17.6 Machine PLC Programming

- (a) PLCs shall be programmable using languages and formats that comply with standard IEC61131-3.
- (b) The PLC programs shall become the property of the Board. The PLC programs shall not have security that prevents the Board from modifying the programs.
- (c) Complete software of the latest (as at completion of commissioning tests) version and revision shall be provided to the Board.

END OF SECTION